Online Appendix

to "Corrective Overoptimism"

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S.A Independent Route Entry and Contestability

When transport carriage is imperfectly competitive, an independently controlled bypass route lowers carrier markups and generates shipper surplus that the entrant does not appropriate. This two-period extension illustrates how diagnostic expectations interact with imperfect route contestability and Mankiw and Whinston (1986) business stealing.

Operating equilibrium.— At date t, a chartered entrant may sink construction cost $F_t^I > 0$ to open an independent bypass at date $t+1$. Once the date-$t+1$ demand intercept v_{t+1} realizes, $n \in \{1, 2\}$ symmetric Cournot transport firms with constant marginal operating cost $c \geq 0$ choose service quantities against linear inverse demand:

$$\tau_{t+1}(D) = v_{t+1} - bD, \qquad b > 0, \quad \text{MC} = c \geq 0. \tag{S.A.1}$$

Writing $a_{t+1} \equiv (v_{t+1} - c)_+$, equilibrium aggregate service D_n, toll τ_n, and per-firm operating profit π_n with n independent routes are

$$D_n = \frac{n}{n+1}\frac{a_{t+1}}{b}, \qquad \tau_n = c + \frac{a_{t+1}}{n+1}, \qquad \pi_n = \frac{a_{t+1}^2}{b(n+1)^2}, \tag{S.A.2}$$

with ownership-adjusted concentration $H_n = 1/n$. Opening an independent bypass ($n = 1 \to 2$) lowers H_n from 1 to 1/2, compresses the markup from $a_{t+1}/2$ to $a_{t+1}/3$, and expands service from

D_1 to D_2 (Figure S.A.1, Panel b), whereas a route extension under incumbent control leaves $n = 1$ and $H_1 = 1$.

Diagnostic beliefs and private vs. social entry.— Suppose $v_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(\mu_{v,t}, \sigma_v^2)$ with $\sigma_v > 0$ and news $\Delta\mu_{v,t} \equiv \mu_{v,t} - \mathbb{E}_{t-1}[v_{t+1}]$. Under diagnosticity $\theta \geq 0$, the perceived mean is $\mu_{v,t}^{\theta} = \mu_{v,t} + \theta\Delta\mu_{v,t}$, and with standardized gap $z_t^{\theta} \equiv (\mu_{v,t}^{\theta} - c)/\sigma_v$, the perceived second moment of positive net demand is

$$M_t^{\theta} \equiv \mathbb{E}_t^{\theta}\left[(v_{t+1} - c)_+^2\right] \tag{S.A.3}$$

$$= \sigma_v^2\Big[\{1 + (z_t^{\theta})^2\}\Phi_N(z_t^{\theta}) + z_t^{\theta}\phi_N(z_t^{\theta})\Big], \tag{S.A.4}$$

which is strictly increasing in z_t^{θ}. For discount factor $\beta \in (0, 1)$, the duopoly entrant's perceived net present value is

$$V_t^{P,\theta} = \frac{\beta M_t^{\theta}}{9b} - F_t^I \geq 0. \tag{S.A.5}$$

Aggregate transport-market surplus with n independent carriers is

$$S_n(v) = \int_0^{D_n} (v - c - bq)\, dq = \frac{n(n+2)}{2(n+1)^2}\frac{(v-c)_+^2}{b}, \tag{S.A.6}$$

so moving from monopoly to duopoly raises direct transport surplus by $5(v-c)_+^2/(72b)$. Letting Γ_t^{ext} denote the discounted external spatial-reallocation and knowledge-diffusion gains not captured by route operators, objective social net present value is

$$V_t^S = \beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} - F_t^I. \tag{S.A.7}$$

Since expected entrant profit $8M_t^0/(72b)$ exceeds the direct transport-surplus gain $5M_t^0/(72b)$ by the Mankiw and Whinston (1986) business-stealing transfer $3M_t^0/(72b) = M_t^0/(24b)$ from the incumbent, rational entry is socially insufficient whenever external spillovers satisfy $\Gamma_t^{\text{ext}} > M_t^0/(24b)$. Diagnostic optimism ($\Delta\mu_{v,t} > 0$) corrects rational under-entry when

$$\frac{\beta M_t^0}{9b} < F_t^I < \beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} \quad \text{and} \quad F_t^I \leq \frac{\beta M_t^{\theta}}{9b}, \tag{S.A.8}$$

and induces socially excessive duplication when

$$\beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} < F_t^I \leq \frac{\beta M_t^{\theta}}{9b}. \tag{S.A.9}$$

Figure S.A.1 illustrates the corrective threshold (Panel a) and the post-entry expansion in service and Marshallian surplus (Panel b), consistent with empirical evidence on carrier market power and network entry (Brancaccio et al., 2020; Caoui and Steck, 2025; Hausman, 2025; Preonas, 2024).

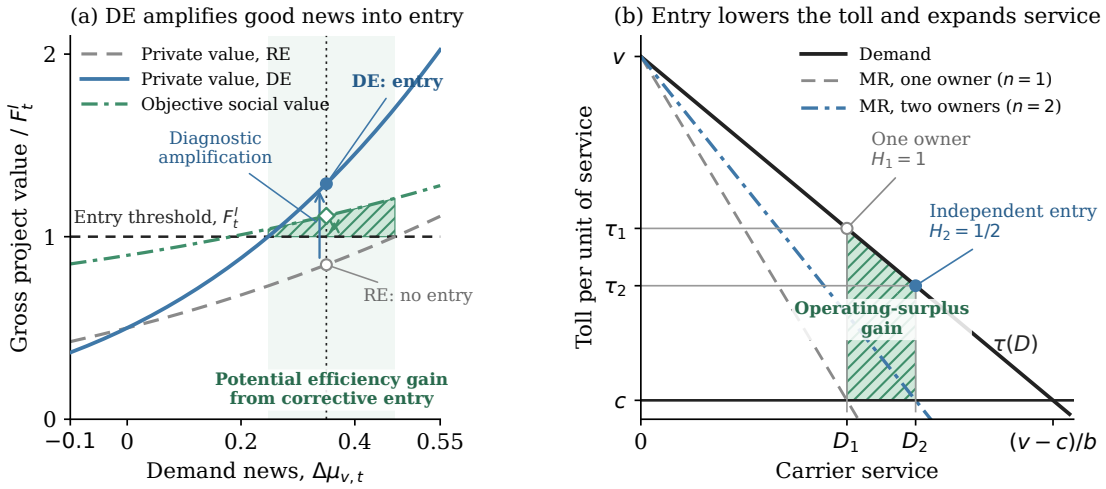


Figure S.A.1: Diagnostic Beliefs and Route Entry

Notes: Panel (a) sets $\beta = 0.9$, $c = 0$, $b = 1$, $\sigma_v = 1$, $\mathbb{E}_{t-1}[v_{t+1}] = 0$, $\theta = 0.9$, $\Gamma_t^{\text{ext}} = 0.065$, and $F_t^I = 0.1$. Panel (b) sets $v - c = b = 1$.

S.B Historical Data and Spatial Aggregation

National and regional occupational series use a common Primary–Secondary–Tertiary (PST) classification. National male (age-15-and-over) occupational shares for England and Wales come from Shaw-Taylor et al. (2019, Table 11) at the benchmark years 1710, 1761, 1817, 1851, 1861, 1871, and 1881, combined with the pre-census trajectory of Keibek (2017, Table 18) interpolated in additive log-ratio differences between adjacent CAMPOP anchors to form the twelve-date panel

$$\mathcal{Y}_O = \{1710, 1721, 1741, 1761, 1781, 1801, 1817, 1841, 1851, 1861, 1871, 1881\}.$$

Primary employment excludes mining; secondary industry includes manufacturing, mining, and construction; and tertiary employment comprises dealers, sellers, transport, and communications (with the sum of these three categories forming the denominator). Because Keibek (2017) classifies mining within primary employment, mining counts are transferred from primary to secondary before normalizing county shares or aligning to the CAMPOP national totals.

Regional occupational shares come from the county reconstructions in Keibek (2017) on fixed 1831 ancient-county boundaries (observed at 1721, 1741, 1761, 1781, 1801, 1817, 1841, and 1851, with 1710 interpolated in additive log ratios between 1701 and 1721) and the dedicated London benchmarks of Field and Shaw-Taylor (2019) (c. 1711, c. 1751, c. 1817, 1851, and 1881, mapped to the fixed 1851 London registration-county footprint). Log-interpolated county populations from Wrigley (2011, Table A2.6) and London population anchors supply the fixed exposure weights N_{by} that convert county and metropolitan occupational shares into relative employment mass. Figures S.B.1 and S.B.2 display the resulting primary and secondary employment shares across ancient counties.

To project county and London shares onto the twenty-five equal-latitude bands $b \in \{1, \ldots, 25\}$ on $\ell \in [0, 1]$, let B_{cb} denote the share of band b's land area lying in county c, $p_{sby} \in [0, 1]$ the share of sector $s \in \{A, M, T\}$ in band b and year y, and N_{by} the reconstructed band employment mass (with normalized weights $\omega_{by} \equiv N_{by} / \sum_{b'=1}^{25} N_{b'y}$ and county weights $W_{cy} \equiv \sum_{b=1}^{25} B_{cb} N_{by}$). The

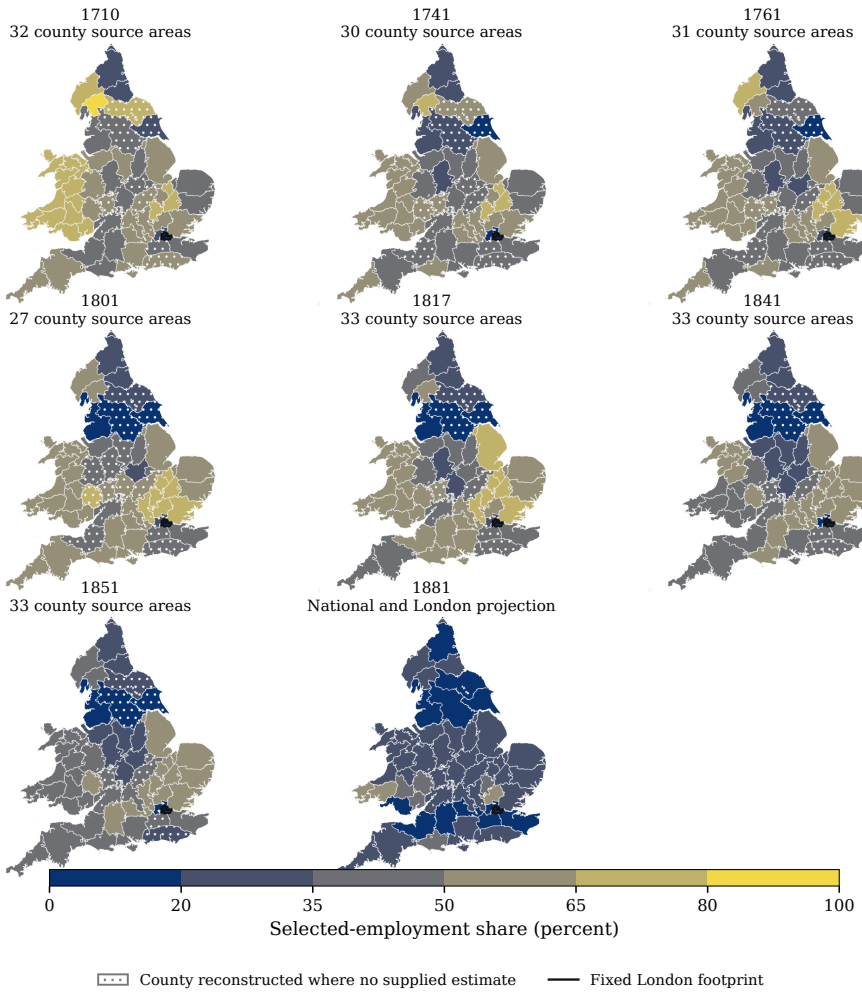


Figure S.B.1: Primary Employment Shares, 1710–1881

Notes: County primary shares through 1851 combine Keibek (2017) and Field and Shaw-Taylor (2019) (London, black outline), and 1881 projects the reconstructed latitude-band surface. Dotted counties indicate reconstructed values.

implied county share is

$$\widehat{p}_{csy} = \frac{\sum_{b=1}^{25} B_{cb} N_{by} p_{sby}}{\sum_{b=1}^{25} B_{cb} N_{by}}.$$

At each pre-1881 date $y \in \mathcal{Y}_O \setminus \{1881\}$, the twenty-five band shares $\{p_{sby}\}$ solve the constrained quadratic program

$$\min_{\{p_{sby} \in [0,1]\}} \sum_{s \in \{A,M,T\}} \left\{ \sum_{c} W_{cy} \left(\widehat{p}_{csy} - p_{csy}^{\text{obs}}\right)^2 + \lambda_{\text{occ}} \sum_{b=2}^{24} \left(p_{s,b+1,y} - 2p_{sby} + p_{s,b-1,y}\right)^2 \right\}$$

subject to $\sum_{s \in \{A,M,T\}} p_{sby} = 1$ for each band b and exact national occupational accounting $\sum_{b=1}^{25} \omega_{by} p_{sby} = P_{sy}^{\text{nat}}$, with spatial roughness weight $\lambda_{\text{occ}} = 0.05$. At the post-1851 census dates $y \in \{1861, 1871, 1881\}$, where ancient-county occupational tables are unavailable outside London, the band surface $\{p_{sby}\}$ updates the 1851 band profile in additive log ratios to match the observed London share $p_{\text{Lon},s,y}^{\text{obs}}$ on the metropolitan bands and the CAMPOP national totals P_{sy}^{nat}, and

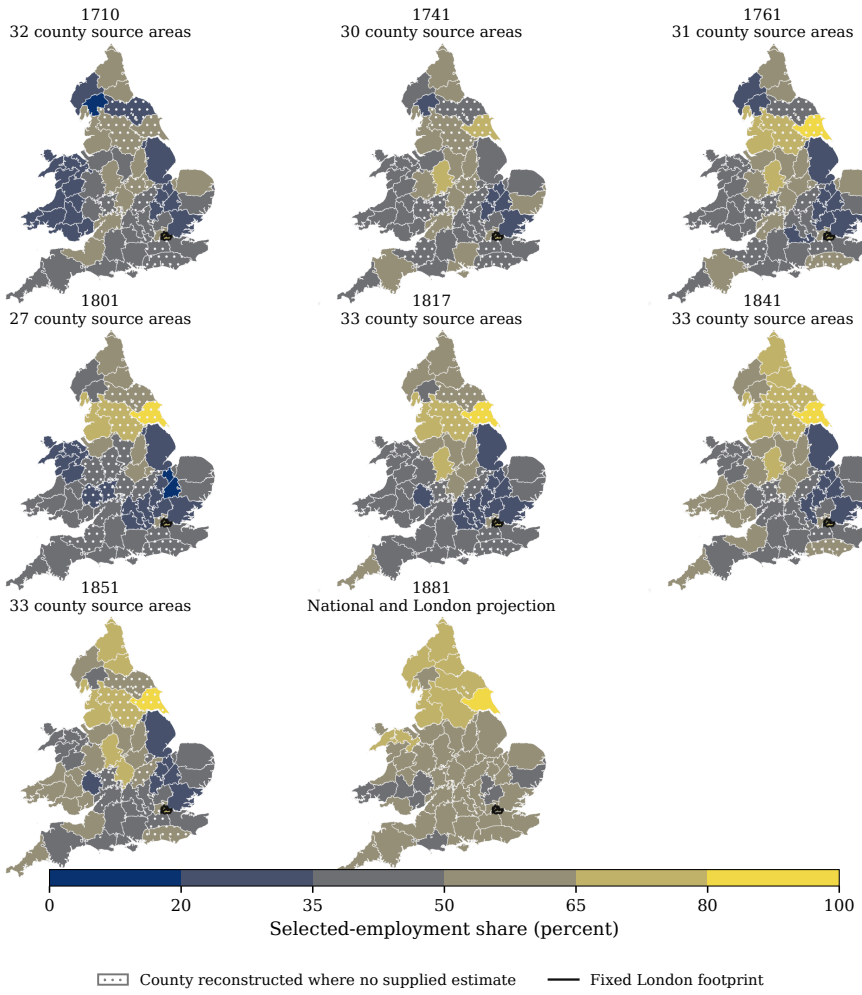


Figure S.B.2: Secondary Employment Shares, 1710–1881

Notes: County secondary shares through 1851 combine Keibek (2017) and Field and Shaw-Taylor (2019) (London, black outline), and 1881 projects the reconstructed latitude-band surface. Dotted counties indicate reconstructed values.

the 1881 county maps in Figures 3, S.B.1, and S.B.2 evaluate the area- and population-weighted projection $\widehat{p}_{c,s,1881} = \sum_{b=1}^{25} B_{cb} N_{b,1881} p_{s,b,1881} / \sum_{b=1}^{25} B_{cb} N_{b,1881}$ (holding London at $p^{\text{obs}}_{\text{Lon},s,1881}$). Each 25-band profile is prolonged piecewise constantly to the 496 interior cells $\{3, \ldots, 498\}$ of the $N = 500$ model grid supported by county centroids (with the four extreme coastal cells $\{1, 2, 499, 500\}$ outside the county-centroid latitude range assigned by nearest-neighbor boundary continuation; Figures S.B.3–S.B.4). The same fixed historical weights N_{by} are applied to data and simulated allocations. Figure S.B.5 reports the complementary 1851 and 1881 registration-subdistrict resident population profiles across the twenty-five latitude bands.

Robustness of regional local projections.— Table S.B.1 reports eight specification checks on the regional local projections in Panel A of Table 1. Row 1 reports two-way standard errors clustered by latitude band and year on the baseline TWFE specification, confirming that cross-corridor spatial correlation does not drive inference (the same one-year reversal holds in the national series, $\rho_1 = -0.325$, SE $= 0.074$). Row 2 replaces leave-one-out shift-share instruments with leave-corridor-out shifts excluding the two adjacent bands on each side ($|k - b| \geq 3$) so trunk projects

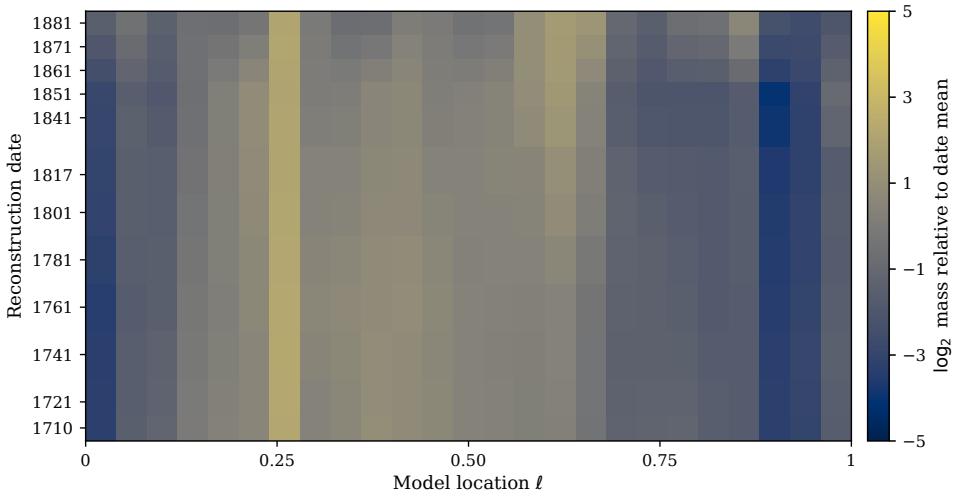


Figure S.B.3: Employment across the Model Interval

Notes: Within-date normalized adult-male employment mass across the twelve benchmark dates using county population weights from Wrigley (2011).

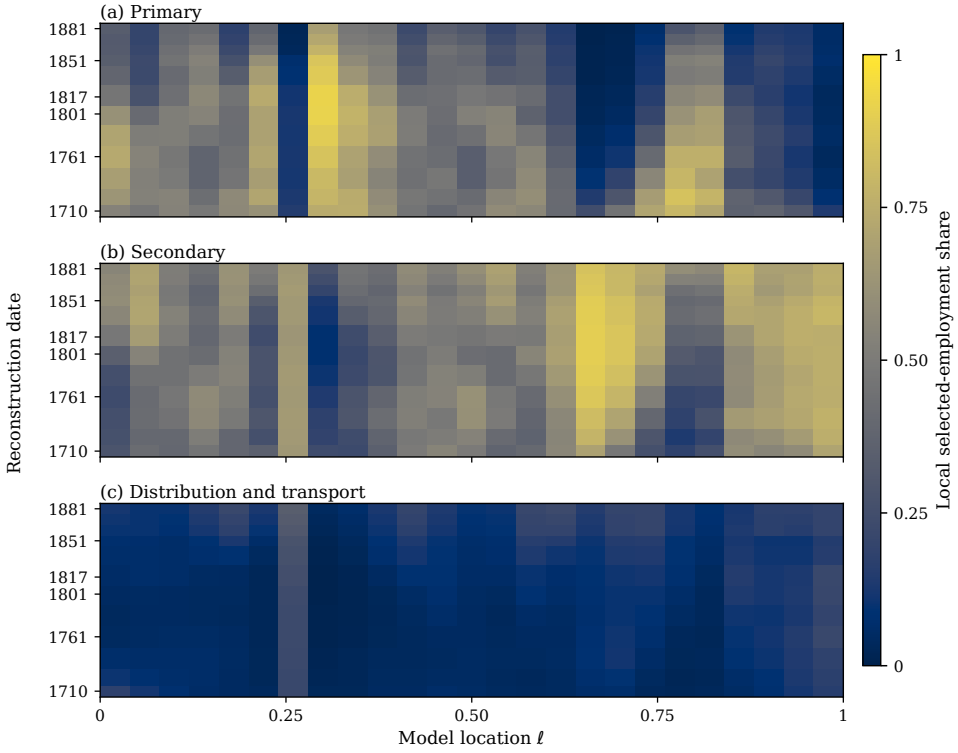


Figure S.B.4: Occupational Composition across the Model Interval

Notes: Primary, secondary, and distribution/transport employment shares across the twenty-five latitude bands at the twelve benchmark dates.

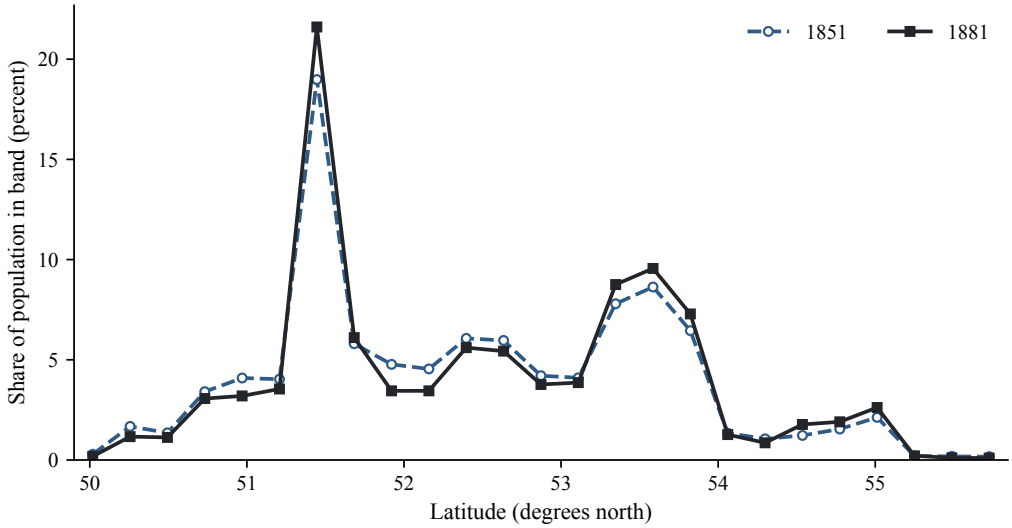


Figure S.B.5: Population across the Model Interval, 1851 and 1881

Notes: CAMPOP registration-subdistrict census population allocated by polygon area to the twenty-five model latitude bands (49.8998–55.8442°N).

spanning neighboring bands cannot correlate the instrument with local residuals. Row 3 detrends log addition rates via a one-sided backward-looking five-year moving average, $c_{b,t}^{1s} = \ln(10^{-3} + r_{b,t}^{I}) - \frac{1}{5}\sum_{j=1}^{5} \ln(10^{-3} + r_{b,t-j}^{I})$, which uses no future realizations after date t and shows positive one-year project-completion momentum ($\rho_1^{1s} = +0.159$) followed by significant reversals at $h = 3, 4, 5$. Row 4 estimates cumulative changes $\Delta_h \ln(10^{-3} + r_{b,t}^{I})$ on unfiltered log addition rates with band and year fixed effects ($\beta_h^{\text{diff}} \in [-0.834, -0.632]$), capturing rapid partial mean reversion ($1 + \beta_h^{\text{diff}} \in [0.166, 0.368]$), whereas level projections on detrended cycles capture drops below local trend. Rows 5 and 6 replace $\log(10^{-3}+r_{b,t}^{I})$ with the level addition rate $r_{b,t}^{I}$ and $\text{arsinh}(100\, r_{b,t}^{I})$, and Rows 7 and 8 split the sample at 1830 into the pre-railway (1710–1829) and railway (1830–1880) eras.

Table S.B.1: Alternative Local-Projection Specifications for Regional Infrastructure Additions

Specification	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
1. Baseline TWFE (two-way band & year clustered SE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.022)	(0.024)	(0.023)	(0.023)	(0.022)
2. Leave-corridor-out 2SLS ($\lvert k-b\rvert \geq 3$)	−0.240	−0.220	−0.096	−0.108	0.003
	(0.050)	(0.036)	(0.043)	(0.048)	(0.036)
3. Strictly one-sided 5-year moving-average filter $c^{1s}_{b,t}$	0.159	−0.017	−0.115	−0.169	−0.196
	(0.020)	(0.016)	(0.019)	(0.018)	(0.014)
4. Unfiltered cumulative change $\Delta_h \ln(10^{-3} + r^I_{b,t})$	−0.632	−0.735	−0.785	−0.806	−0.834
	(0.035)	(0.033)	(0.036)	(0.036)	(0.033)
5. Untransformed level addition rate $\mathrm{HP}_{6.25}[r^I_{b,t}]$	−0.138	−0.284	−0.136	−0.056	0.052
	(0.017)	(0.026)	(0.015)	(0.016)	(0.046)
6. Inverse hyperbolic sine $\mathrm{HP}_{6.25}[\mathrm{arsinh}(100\, r^I_{b,t})]$	−0.184	−0.226	−0.142	−0.050	−0.001
	(0.020)	(0.023)	(0.020)	(0.017)	(0.019)
7. Pre-railway turnpike and canal era (1710–1829)	−0.255	−0.201	−0.101	−0.027	−0.004
	(0.020)	(0.022)	(0.021)	(0.023)	(0.023)
8. Railway era (1830–1880)	0.067	−0.342	−0.274	−0.034	−0.082
	(0.050)	(0.045)	(0.028)	(0.035)	(0.036)

Notes: All specifications include latitude-band and year fixed effects on the 25-band regional transport panel. Standard errors in parentheses are clustered by latitude band (except Row 1, which clusters two-way by band and year).

S.C Stochastic Primitives

This appendix derives the properties of the two stochastic primitives of Section III that the main text uses: the bounded law of aggregate sectoral productivity gaps, with the diagnostic one-step law it implies, and the spatially correlated discovery field, which keeps the knowledge frontier finite.

S.C.1 Bounded sectoral shocks and the diagnostic law

Let $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$ with $|\rho_s| < 1$, and let $\varphi_{\boldsymbol{\Sigma}}$ be the centered Gaussian density with innovation covariance $\boldsymbol{\Sigma} = \mathbf{D}_z \boldsymbol{\Omega}_z \mathbf{D}_z$, where $\mathbf{D}_z = \mathrm{diag}(\sigma_A, \sigma_M, \sigma_T)$ and $\boldsymbol{\Omega}_z$ is the correlation matrix with off-diagonal entries $(\varrho_{AM}, \varrho_{AT}, \varrho_{MT})$, parameterized via the partial correlation $\varrho_{MT|A} \in (-1,1)$ as $\varrho_{MT} = \varrho_{AM}\varrho_{AT} + \varrho_{MT|A}\sqrt{(1-\varrho^2_{AM})(1-\varrho^2_{AT})}$ so that $\boldsymbol{\Omega}_z$ is positive definite and the diagonal entries of $\boldsymbol{\Sigma}$ are $\Sigma_{ss} = \sigma_s^2$. Because $\boldsymbol{\rho}$ is diagonal, the untruncated stationary standard deviation of z_s is $\nu_s = \sigma_s/\sqrt{1-\rho_s^2}$ whatever the cross-sector correlations. The compact symmetric support box, box probability, and truncated conditional mean are

$$\begin{aligned} \mathcal{B} &= \prod_{s\in\{A,M,T\}} [-c\nu_s, c\nu_s], \quad \Phi_{\mathcal{B}}(\mathbf{m}) = \int_{\mathcal{B}} \varphi_{\boldsymbol{\Sigma}}(\mathbf{u}-\mathbf{m})\, d\mathbf{u}, \\ M(\mathbf{m}) &= \frac{\int_{\mathcal{B}} \mathbf{u}\, \varphi_{\boldsymbol{\Sigma}}(\mathbf{u}-\mathbf{m})\, d\mathbf{u}}{\Phi_{\mathcal{B}}(\mathbf{m})}, \end{aligned} \tag{S.C.1}$$

with $c = 3$ in the benchmark and $c = 4$ reserved for robustness, and the physical kernel in (9) is $f(\mathbf{z}' \mid \mathbf{z}) = \varphi_{\boldsymbol{\Sigma}}(\mathbf{z}' - \boldsymbol{\rho}\mathbf{z})/\Phi_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z})$ for $\mathbf{z}' \in \mathcal{B}$. Since f is continuous and strictly positive on the compact set $\mathcal{B}\times\mathcal{B}$, it is bounded below by some $\underline{f} > 0$, so $\Pr(\mathbf{z}_{t+1} \in E \mid \mathbf{z}_t) \geq \underline{f}\,|E|$ for every Borel set $E \subseteq \mathcal{B}$ with Lebesgue measure $|E|$. By this Doeblin minorization, f has a unique stationary distribution π on $\mathcal{B}$, approached geometrically from every initial state, and the normalizer in (18),

$$\mathbb{E}_\pi[e^{z_s}] = \int_{\mathcal{B}} e^{z_s}\, \pi(d\mathbf{z}), \qquad s \in \{A, M, T\}, \tag{S.C.2}$$

gives $e^{z_{st}}/\mathbb{E}_\pi[e^{z_s}]$ unit stationary physical mean. For any truncated Gaussian law $\mathbf{z} \sim \mathrm{TN}_{\mathcal{B}}(\mathbf{m}, \boldsymbol{\Sigma})$, completing the square in $\mathbf{u}'\mathbf{e}_i - \frac{1}{2}(\mathbf{u}-\mathbf{m})'\boldsymbol{\Sigma}^{-1}(\mathbf{u}-\mathbf{m})$ gives the conditional mean of effective productivity, with $\mathbf{m} = \boldsymbol{\rho}\mathbf{z}_t$ under physical beliefs and $\mathbf{m} = \mathbf{m}_t^\theta$ under diagnostic beliefs:

$$\mathbb{E}\left[\frac{A_i e^{z_i}}{\mathbb{E}_\pi[e^{z_i}]}\,\middle|\,\mathbf{m}\right] = \frac{A_i}{\mathbb{E}_\pi[e^{z_i}]}\, e^{m_i + \frac{1}{2}\sigma_i^2}\, \frac{\Phi_{\mathcal{B}}(\mathbf{m} + \boldsymbol{\Sigma}\mathbf{e}_i)}{\Phi_{\mathcal{B}}(\mathbf{m})}. \tag{S.C.3}$$

For the diagnostic law, the box normalizers $\Phi_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t)$ and $\Phi_{\mathcal{B}}(\boldsymbol{\rho}\bar{\mathbf{z}}_t)$ in the representativeness ratio Λ_t do not depend on the future state and cancel from (10). The remaining kernel $\varphi_{\boldsymbol{\Sigma}}(\mathbf{u} - \boldsymbol{\rho}\mathbf{z}_t)^{1+\theta}\varphi_{\boldsymbol{\Sigma}}(\mathbf{u} - \boldsymbol{\rho}\bar{\mathbf{z}}_t)^{-\theta}$ has log-quadratic term $-\frac{1}{2}\mathbf{u}'\boldsymbol{\Sigma}^{-1}\mathbf{u}$, because the exponents $1+\theta$ and $-\theta$ sum to one, and linear term $\mathbf{u}'\boldsymbol{\Sigma}^{-1}[(1+\theta)\boldsymbol{\rho}\mathbf{z}_t - \theta\boldsymbol{\rho}\bar{\mathbf{z}}_t]$. Restricted to $\mathcal{B}$, it is therefore a truncated Gaussian with covariance $\boldsymbol{\Sigma}$ and location $\boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \bar{\mathbf{z}}_t)$, which is (11).

S.C.2 Spatially correlated discoveries and frontier moments

If discoveries were drawn independently across sites, any interval of active research would contain infinitely many independent Pareto draws, which are unbounded almost surely, and the spatial supremum (38) would be infinite. The discovery field (23) keeps the Bernoulli–Pareto law at every site but draws discoveries from Gaussian fields with continuous sample paths, which bounds the frontier and its moments.

LEMMA S.C.1: *Fix sector $i \in \{A, M\}$, bounded positive inherited knowledge A_{it}, pre-discovery plans (e_{it}, ϕ_{it}), and discoveries $G_{it}(\ell) = G_i(\phi_{it}(\ell); \ell, \omega)$ generated by (23) with correlation length $d_0 > 0$. Then the frontier $F_{it}(\ell)$ in (38) is finite almost surely, and for every $0 < b < a_i$ and $F_{it}^0(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|}A_{it}(r)$,*

$$\mathbb{E}_t[F_{it}(\ell)^b] < \infty, \tag{S.C.4}$$

$$\mathbb{E}_t[F_{it}(\ell)^b] = [F_{it}^0(\ell)]^b + b\int_{F_{it}^0(\ell)}^{\infty} v^{b-1}\Pr_t(F_{it}(\ell) > v)\, dv. \tag{S.C.5}$$

For $a_i > 1$, next-period expected knowledge under (39) satisfies

$$\mathbb{E}_t[A_{i,t+1}(\ell)] = e^{-\lambda_t(\ell)}A_{it}(\ell)\left[1 + \frac{e_{it}(\ell)\phi_{it}(\ell)}{a_i - 1}\right] + (1 - e^{-\lambda_t(\ell)})\mathbb{E}_t[F_{it}(\ell)]. \tag{S.C.6}$$

Proof. Because X_{it}^g has continuous sample paths on the compact set $[0,1]$, $\overline{X} = \sup_{r\in[0,1]}[-X_{it}^g(r)]$ is finite almost surely, and the Borell–TIS inequality (Adler et al., 2015, Theorem 2.6.1) with unit variance gives $\bar{\mu} = \mathbb{E}[\overline{X}] < \infty$ and $\Pr(\overline{X} \geq \bar{\mu} + x) \leq e^{-x^2/2}$ for $x > 0$. Pathwise, $1 \leq G_{it}(r) \leq \Phi(X_{it}^g(r))^{-1/a_i} \leq \Phi(-\overline{X})^{-1/a_i}$, and the Gaussian tail bound $\Phi(-x) \geq c_\Phi(1+x)^{-1}e^{-x^2/2}$ ($x \geq 0$, $c_\Phi > 0$) gives $\sup_{r\in[0,1]} G_{it}(r)^b \leq C(1 + \overline{X}_+)^{b/a_i}\exp\{(b/(2a_i))\overline{X}_+^2\}$, which is integrable whenever $b < a_i$. Since $F_{it}(\ell) \leq \sup_r A_{it}(r)\sup_r G_{it}(r)$, the frontier is finite almost surely and (S.C.4) holds. Because $G_{it} \geq 1$ implies $F_{it} \geq F_{it}^0$, the layer-cake identity gives (S.C.5). Finally, $\mathbb{E}_t[G_{it}(\ell) - 1] = \phi_{it}(\ell)/(a_i - 1)$ under (21)–(22), and taking expectations in $A_{i,t+1} = e^{-\lambda_t}A_{it}^+ + (1 - e^{-\lambda_t})F_{it}$ with (37) and $\mathbb{E}_t[F_{it}] < \infty$ ($b = 1 < a_i$) yields (S.C.6). □

S.D Estimation Design, Moment Construction, and Covariance

This section details the parameterization, empirical observation maps, moment formulas, and weighting matrix used in the Simulated Method of Moments (SMM) estimation of Section IV.C.

S.D.1 Estimated and fixed parameters

Under diagnostic expectations, the SMM criterion (77) jointly estimates $p = 26$ free structural parameters: $\theta \geq 0$, the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters (η, γ, a_A, a_M, f_A, f_M, χ_A, χ_M, ω_E, δ, λ_0, α_T, κ_0, r_A, m_I, χ_I), and the 9 aggregate sectoral shock parameters

$$\boldsymbol{\psi}^z = (\rho_A, \rho_M, \rho_T, \sigma_A, \sigma_M, \sigma_T, \varrho_{AM}, \varrho_{AT}, \varrho_{MT|A})', \tag{S.D.1}$$

while the manufacturing wedge scale $\bar{\psi}_M = 0.251$ (SE $= 0.032$) is calibrated from the gross-operating-surplus identity (76) (Table 5). Under rational expectations, the model is re-estimated subject to $\theta = 0$ ($p = 25$ free parameters). Table 4 summarizes the identifying margins and Table 3 lists the fixed parameters.

The initial 1710 log-productivity profiles $\ln A_{i,1710}(\ell)$ ($i \in \{A, M\}$) use a five-knot natural cubic spline basis on $\{0, 1/4, 1/2, 3/4, 1\}$ with four shape parameters per sector and zero second derivatives at $\ell \in \{0, 1\}$. Goods units normalize $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$ for $i \in \{A, M\}$, and at each candidate parameter vector the eight spline parameters are concentrated out on the 1710 occupational cross-section. Nominal units fix $p_A(1, t) = 1$, household mass starts at $\bar{L}_{1710} = 100$ and evolves via (8), and the 1710 multimode transport stock has cross-band mean one. Every candidate solves the conditional dated equilibrium and the Markov perfect county investment equilibrium on the $N = 500$ grid across $J = 10$ counties, matching the $M = 30$ moments in Table 6 ($M - p = 4$ overidentifying degrees of freedom under DE and 5 under RE).

S.D.2 Observation maps and empirical roles

At each calendar date y, sectoral employment aggregates over grid cells of width $\Delta\ell_q$ (Replication Appendix R.A.3.1) and national occupational shares are formed from predetermined production and carrier hiring:

$$\begin{aligned} \mathcal{L}_A(y) &= \sum_{q=1}^{N} \Delta\ell_q e_{Aq}(y) L_{Aq}(y), \quad \mathcal{L}_M(y) = \sum_{q=1}^{N} \Delta\ell_q e_{Mq}(y) L_{Mq}(y), \\ \mathcal{L}_T(y) &= \sum_{q=1}^{N} \Delta\ell_q L_{Tq}(y), \qquad\qquad \pi^{\text{occ}}_{o,y} = \frac{\mathcal{L}_o(y)}{\sum_{o' \in \{A,M,T\}} \mathcal{L}_{o'}(y)}, \quad o \in \{A, M, T\}, \end{aligned} \tag{S.D.2}$$

satisfying $\sum_{o \in \{A,M,T\}} \mathcal{L}_o(y) = \sum_{q=1}^{N} \Delta\ell_q n_q(y) = \bar{L}_y$ since innovation is goods-financed (25) and hires no separate research headcount. National occupational shares aggregate the model domain, whereas dated spatial covariances aggregate pure-site employment onto the 496 supported cells of the twenty-five latitude bands. Table S.D.1 summarizes the empirical series and their quantitative roles.

Observed 1710 transport capacity initializes $T_0(\ell)$, whereas post-1710 stocks enter only as empirical moment targets. Because the source-bootstrap covariance matrix conditions on the historical spatial and demographic reconstructions, the asymptotic standard errors and Hansen (1982) J-statistics are interpreted conditionally on those reconstructed surfaces.

S.D.3 Complete historical profiles

To capture both the timing and steepness of the Industrial Revolution, the targeted moments summarize the complete temporal trajectories of the national-account, occupational, and transport-stock series plotted in Figure S.D.1. For an n-date series $z = (z_1, \ldots, z_n)'$, let $H_n \in \mathbb{R}^{(n-1)\times n}$ have

Table S.D.1: Empirical Objects and Their Quantitative Roles

Object	Transformation	Role in estimation and validation
Factor payments	Decade-specific nominal wage–output ratios and agricultural rent–wage ratios	Set production labor elasticities (α_A, α_M) externally (Table 3) and discipline manufacturing wedge scale $\bar{\psi}_M$
Annual national accounts	Linear trends in log real GDP, $\ln(Y_M/Y_A)$, and $\ln(p_M/p_A)$ over complete profiles; aggregate real wage $w/\mathcal{P}$	Discipline aggregate growth, structural transformation, and relative prices (Table 6, Panel A), with the real wage path $w/\mathcal{P}$ reserved for untargeted validation (Figure 7, Panel b)
Sectoral activity cycles	Ravn–Uhlig ($\lambda_{\text{HP}} = 6.25$) HP-filtered variances, autocovariances, and cross-covariances	Discipline aggregate shock persistence $\boldsymbol{\rho}$ and innovation covariance $\boldsymbol{\Sigma}$ (Panel D)
Occupation surfaces	Linear trends in π_A^{occ} and π_T^{occ} and mean dated spatial covariances $(C_{l,y}, C_{e,y}, C_{N,y}, V_{T,y}^{\text{occ}})$	Discipline initial productivity $A_i(\ell, 1710)$, Stone–Geary (η, γ), and spatial sorting (Panels A, C)
Multimode transport stock	Linear and quadratic trends, 1830–1850 growth, post-1850 slope change, spatial log-variance $V_{T,y}$, and investment dynamics	Discipline construction cost scale m_I, adjustment convexity χ_I, and diagnosticity θ (Panels B, E)
Manufacturing & energy records	Normalized fire-insurance wedge exposure $\omega_M(\ell)$ and coalfield cost profile $\xi(\ell)$	Supply predetermined spatial wedge and energy-access schedules
Patents, railways & carrier accounts	Geocoded patent–railway panels, working accounts, and route concentration	Reserved for external checks of transport-mediated diffusion and carriage outside the estimation criterion

orthonormal rows orthogonal to $\mathbf{1}_n$:

$$H_n \mathbf{1}_n = 0, \qquad H_n H_n' = I_{n-1}, \qquad H_n' H_n = I_n - \frac{\mathbf{1}_n \mathbf{1}_n'}{n}. \tag{S.D.3}$$

Because $H_n' H_n z = z - \bar{z}\mathbf{1}_n$, retaining $H_n z \in \mathbb{R}^{n-1}$ preserves the complete demeaned trajectory while remaining invariant to the choice of orthonormal basis under covariance rotation. The three national-account profiles are log aggregate real GDP (1710–1881, $n = 172$), log industrial output relative to agriculture $\ln(Y_M/Y_A)$ (1710–1881, $n = 172$; Broadberry et al. 2022 linked at 1870 to Feinstein 1972, Table 54), and log relative industrial price $\ln(p_M/p_A)$ (1710–1870, $n = 161$), whose OLS linear trends on calendar time $w_{\text{lin}} = (y - \bar{y})/\sum_k (y_k - \bar{y})^2$ form the first three moments in Panel A of Table 6.

S.D.4 Occupations and spatial reallocation

Evaluating (S.D.2) across the twelve reconstruction dates $y \in \mathcal{Y}_O$ supplies the national occupational profiles $(\pi_{A,y}^{\text{occ}}, \pi_{T,y}^{\text{occ}})$ (spanning the three-sector simplex since $\sum_{o \in \{A,M,T\}} \pi_{o,y}^{\text{occ}} = 1$), whose OLS linear trends over 1710–1881 complete Panel A of Table 6 (5 moments in total). On the 496 supported cells d carrying the twenty-five band occupational shares $\pi_{o,d,y}$, with equal-cell mean $\langle \cdot \rangle$, standardized latitude l_d, predetermined coalfield energy exposure e_d, and adjacent-cell averaging matrix A, the four dated spatial moments at each $y \in \mathcal{Y}_O$ are

$$\begin{aligned} C_{l,y} &= \big\langle l_d \big(\pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle\big) \big\rangle, \quad & C_{e,y} &= \big\langle e_d \big(\pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle\big) \big\rangle, \\ C_{N,y} &= \big\langle \widetilde{\pi}_{M,d,y} (A \widetilde{\pi}_{M,\cdot,y})_d \big\rangle, & V_{T,y}^{\text{occ}} &= \big\langle \big(\pi_{T,d,y} - \langle \pi_{T,\cdot,y} \rangle\big)^2 \big\rangle, \end{aligned} \tag{S.D.4}$$

where $\widetilde{\pi}_{M,d,y} = \pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle$, and their sample means across $y \in \mathcal{Y}_O$ form the 4 moments in Panel C of Table 6.

S.D.5 Infrastructure accumulation and dispersion

Aggregating installed route capacity $T_t(\ell)$ onto the twenty-five latitude bands $h \in \{1, \dots, 25\}$ at each annual date $y \in \{1710, \dots, 1881\}$ yields the national mean stock $\overline{T}_y$ and cross-band log-stock variance $V_{T,y}$:

$$\overline{T}_y = \frac{1}{25}\sum_{h=1}^{25} T_{h,y}, \qquad V_{T,y} = \frac{1}{25}\sum_{h=1}^{25}\left(\log T_{h,y} - \frac{1}{25}\sum_{k=1}^{25}\log T_{k,y}\right)^2. \tag{S.D.5}$$

In Panel B of Table 6 (5 moments), the trajectory of $\ln \overline{T}_y$ is summarized by four linear functionals of $H_{172}\ln \overline{T}$—the full-sample OLS linear trend, the orthogonal quadratic trend $2\,(q'\ln \overline{T})/(q'q)$ on century-rescaled calendar time $\tilde{t}_y = (y - \bar{y})/100$ (with $q_y = \tilde{t}_y^2 - \overline{\tilde{t}^2}$ orthogonalized to $\tilde{t}$), the 1830–1850 OLS growth rate, and the post-1850 slope change (the difference between the 1850–1881 and 1830–1850 OLS growth rates)—together with the full-sample mean of $V_{T,y}$.

S.D.6 Investment dynamics and local returns

From the annual national commitment rate r_y^I and its HP-filtered log cycle c_y^I,

$$r_y^I = \frac{\overline{T}_{y+1} - \overline{T}_y}{\overline{T}_y}, \qquad c_y^I = \mathrm{HP}_{6.25}\left[\log(10^{-3} + r_y^I)\right], \tag{S.D.6}$$

the 7 investment-dynamics moments in Panel E of Table 6 are the standard deviation $\sigma(c_t^I) = \mathrm{RMS}(c^I)$, the contemporaneous and lag-1 output-cycle slopes (b_0, b_1) from $c_y^I = \alpha + b_0\bar{c}_{Yy} + b_1\bar{c}_{Y,y-1} + u_y$ where $\bar{c}_{Yy} = \frac{1}{3}(c_{Ay} + c_{My} + c_{Ty})$, the first-order autocorrelation ρ_1 ($c_{y+1}^I = \alpha + \rho_1 c_y^I + u_{y+1}$, equivalent to the cumulative change slope $\beta_1^{\mathrm{diff}} = \rho_1 - 1$), the one-year lead output slope b_{lead} from $\bar{c}_{Y,y+1} = \alpha + b_{\mathrm{lead}} c_y^I + u_{y+1}$, and the two-way fixed-effects decadal county capacity-addition slopes (g_1, g_2) on the first and second decadal lags of county population growth across the 41 ancient counties over 1791–1851. Matched agricultural land rents and production-weighted agricultural wages at year y over grid cells of width $\Delta\ell_q$ (Replication Appendix R.A.3.1) satisfy

$$r_y^A = \frac{\sum_{q=1}^N \Delta\ell_q e_{Aq} R_q}{\sum_{q=1}^N \Delta\ell_q e_{Aq}}, \qquad w_y^A = \frac{\sum_{q=1}^N \Delta\ell_q e_{Aq} L_{Aq} w_q}{\sum_{q=1}^N \Delta\ell_q e_{Aq} L_{Aq}}, \tag{S.D.7}$$

deflated by the agricultural producer price index (Clark, 2002).

S.D.7 Sectoral fluctuations

Applying the annual Ravn and Uhlig (2002) Hodrick–Prescott filter ($\lambda_{\mathrm{HP}} = 6.25$, second-difference matrix D_2) identically to historical and simulated annual log sectoral outputs ($n = 172$ years),

$$c_s = \left[I_n - (I_n + 6.25 D_2' D_2)^{-1}\right]\log Y_s, \tag{S.D.8}$$

yields the 9 sectoral cycle moments in Panel D of Table 6—the variances v_s, first-order autocovariances a_s, and cross-sector covariances $k_{ss'}$ for $s, s' \in \{A, M, T\}$:

$$\begin{aligned} v_s &= \frac{1}{n}\sum_{y=1}^{n}(c_{s,y} - \bar{c}_s)^2, \qquad a_s = \frac{1}{n-1}\sum_{y=2}^{n}(c_{s,y-1} - \bar{c}_{s,-})(c_{s,y} - \bar{c}_{s,+}), \\ k_{ss'} &= \frac{1}{n}\sum_{y=1}^{n}(c_{s,y} - \bar{c}_s)(c_{s',y} - \bar{c}_{s'}). \end{aligned} \tag{S.D.9}$$

S.D.8 External benchmarks and covariance

Geocoded British patent and railway opening panels (Berkes et al., 2026; Coll Ardanuy et al., 2021), railway working accounts, and ownership-adjusted carrier concentration indices are reserved as external validation checks outside the SMM objective.

Stacking Panels A–E of Table 6 forms the $M = 30$ empirical moment vector $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ (5 aggregate trends in Panel A, 5 transport-network moments in Panel B, 4 spatial employment covariances in Panel C, 9 HP-filtered sectoral moments in Panel D, and 7 investment-dynamics moments in Panel E). For each moment $k \in \{1, \dots, 23\}$ in Panels A–D with projection weight c_k on the complete profile, $\widehat{\text{SE}}_k = \sqrt{c_k' \widehat{\Sigma}_{\text{prof}} c_k}$, where $\widehat{\Sigma}_{\text{prof}}$ is the Ledoit and Wolf (2004) shrinkage-regularized source-bootstrap covariance matrix from paired time-series resamples of the Broadberry et al. (2022) panel and spatial block-bootstrap resamples of the twenty-five latitude-band histories; for $k \in \{24, \dots, 30\}$ in Panel E, $\widehat{\text{SE}}_k$ comes from the Newey–West HAC covariance of the national investment regressions and the county block-bootstrap covariance.

With inverse-variance weighting $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \dots, 1/\widehat{\text{SE}}_{30}^2)$ and simulated mean $\overline{\mathbf{m}}(\boldsymbol{\psi}) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{m}(\boldsymbol{\psi}; u_b)$, the SMM criterion is

$$Q(\boldsymbol{\psi}) = \left[\overline{\mathbf{m}}(\boldsymbol{\psi}) - \mathbf{m}^{\text{data}}\right]' W \left[\overline{\mathbf{m}}(\boldsymbol{\psi}) - \mathbf{m}^{\text{data}}\right] = \sum_{k=1}^{30} \left(\frac{\overline{m}_k(\boldsymbol{\psi}) - m_k^{\text{data}}}{\widehat{\text{SE}}_k} \right)^2 . \tag{S.D.10}$$

Table S.D.2 reports all $M = 30$ targeted moments, their symbols in (S.D.3)–(S.D.9), empirical estimates $\widehat{m}_k$, standard errors $\widehat{\text{SE}}_k$, weights $W_{kk} = 1/\widehat{\text{SE}}_k^2$, and squared studentized residuals $q_k(\widehat{\Theta}) = W_{kk}(\overline{m}_k(\widehat{\Theta}) - \widehat{m}_k)^2$ under DE and RE. With $J(\widehat{\boldsymbol{\psi}}) = D_{\boldsymbol{\psi}} \overline{\mathbf{m}}(\widehat{\boldsymbol{\psi}})$ and empirical moment covariance $\widehat{\Sigma}_m$, misspecification-robust standard errors follow from the sandwich estimator

$$\widehat{V}_{\boldsymbol{\psi}} = (J'WJ)^{-1} J'W \widehat{\Sigma}_m W J (J'WJ)^{-1}. \tag{S.D.11}$$

S.D.9 Industrial output and the construction account

County construction purchases $I_{jy} = \int_{\mathcal{C}_j} I_y(\ell)\, d\ell$ at cost $C^I_{jy} = \int_{\mathcal{C}_j} p_M(\ell, y) I_y(\ell)\, d\ell$ are a final use of manufactured output deducted from county land rent; manufacturing output $Y_{My} = \int_0^1 Y_M(\ell, y)\, d\ell$ and producer receipts $V_{iy} = \int_0^1 P_i(\ell, y) Y_i(\ell, y)\, d\ell$ ($P_i = p_i - r_i \tau_{j(\ell)}$) already include these investment goods. Following Broadberry et al. (2022), the chain-linked agriculture–manufacturing volume index Y_t^{AM} ($Y_{1710}^{AM} = 1$) weights sectoral quantities within each segment by benchmark value shares V_{ib_s} (with benchmark years $b_s \in \{1710, 1759, 1801, 1841\}$ linked at 1740/41, 1780/81, and 1820/21):

$$\frac{Y_t^{AM}}{Y_{t_s}^{AM}} = \frac{\sum_{i \in \{A,M\}} V_{ib_s}\, Y_{it} / Y_{ib_s}}{\sum_{i \in \{A,M\}} V_{ib_s}\, Y_{it_s} / Y_{ib_s}}. \tag{S.D.12}$$

Figure S.D.2 compares the baseline width-adjusted multimode transport stock (75) with the unadjusted route-mile series.

Table S.D.2: Complete List of Targeted SMM Moments, Standard Errors, and Weights ($M = 30$)

k	Moment description	Symbol	$\widehat{m}_k$	$\widehat{\mathrm{SE}}_k$	$W_{kk} = 1/\widehat{\mathrm{SE}}_k^2$	$q_k(\widehat{\Theta}^{\mathrm{DE}})$	$q_k(\widehat{\Theta}^{\mathrm{RE}})$
	Panel A. Aggregate growth and structural transformation						
1	Log real GDP trend (%/yr)	β_{GDP}	1.473	0.210	22.7	0.052	0.27
2	Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	$\beta_{M/A}$	1.085	0.572	3.05	0.040	0.25
3	Log relative price $\ln(p_M/p_A)$ trend (%/yr)	β_{p_M/p_A}	−0.757	0.616	2.63	0.041	0.18
4	Agricultural employment share trend (pp/yr)	β_{π_A}	−0.146	0.023	1,833.3	0.077	1.61
5	Tertiary employment share trend (pp/yr)	β_{π_T}	0.060	0.007	18,939.4	0.092	0.54
	Panel B. Transport network growth and spatial dispersion						
6	Log transport stock: linear trend (%/yr)	$\beta_{\overline{T},1}$	1.948	0.103	94.3	0.002	3.39
7	Log transport stock: quadratic trend (×100)	$\beta_{\overline{T},2}$	2.923	0.217	21.3	0.018	12.90
8	Log transport stock: 1830–1850 growth (%/yr)	$g_{\overline{T},1830\text{–}50}$	7.470	0.759	1.73	0.006	12.36
9	Log transport stock: post-1850 slope change (%/yr)	$\Delta g_{\overline{T},\text{post-1850}}$	−5.101	0.881	1.29	0.002	8.98
10	Cross-band variance of log transport stock	$\overline{V}_T$	0.469	0.148	45.9	0.047	0.16
	Panel C. Cross-sectional employment covariances						
11	Manufacturing share–latitude covariance	$\overline{C}_l$	0.077	0.026	1,471.2	0.062	0.21
12	Manufacturing share–energy covariance	$\overline{C}_e$	−0.070	0.024	1,804.4	0.061	0.24
13	Spatial autocovariance of manufacturing share	$\overline{C}_N$	0.030	0.006	24,758.3	0.080	0.51
14	Cross-band variance of tertiary share	$\overline{V}_T^{\mathrm{occ}}$	0.0029	0.0011	831,010.2	0.052	0.15
	Panel D. HP-filtered log sectoral output moments ($\times 10^4$)						
15	Variance: Agriculture (A)	v_A	34.79	4.60	0.05	0.050	1.19
16	Variance: Manufacturing (M)	v_M	9.64	1.40	0.51	0.052	3.12
17	Variance: Transport (T)	v_T	5.66	0.93	1.17	0.049	3.44
18	First-order autocovariance: Agriculture (A)	a_A	−7.27	2.41	0.17	0.029	0.05
19	First-order autocovariance: Manufacturing (M)	a_M	−0.49	0.68	2.13	0.043	0.36
20	First-order autocovariance: Transport (T)	a_T	−0.63	0.39	6.52	0.043	0.37
21	Cross-sector covariance: (A, M)	k_{AM}	−2.59	1.43	0.49	0.043	0.82
22	Cross-sector covariance: (A, T)	k_{AT}	−0.55	0.83	1.46	0.036	0.61
23	Cross-sector covariance: (M, T)	k_{MT}	4.74	0.78	1.65	0.072	4.21
	Panel E. Infrastructure investment dynamics						
24	Standard deviation of cyclical investment, $\sigma(c_t^I)$	$\sigma(c_t^I)$	0.431	0.043	537.8	0.008	4.01
25	Investment on current output cycle (b_0)	b_0	2.954	1.431	0.49	0.009	0.39
26	Investment on lagged output cycle (b_1)	b_1	1.585	1.257	0.63	0.004	1.53
27	First-order autocorrelation of investment (ρ_1)	ρ_1	−0.325	0.074	181.4	0.065	3.68
28	Lead output cycle on investment (b_{lead})	b_{lead}	−0.0051	0.0039	65,197.6	0.003	0.16
29	County investment on population growth, lag 1 (g_1)	g_1	−0.162	0.050	400.0	0.324	2.54
30	County investment on population growth, lag 2 (g_2)	g_2	−0.496	0.050	400.0	0.063	49.65
	Subtotal: Panels A–C ($k = 1, \ldots, 14$)					0.63	41.74
	Subtotal: Panel D ($k = 15, \ldots, 23$)					0.42	14.16
	Subtotal: Panel E ($k = 24, \ldots, 30$)					0.48	61.96
	Total weighted SMM criterion $Q(\widehat{\Theta}) = \sum_{k=1}^{30} q_k(\widehat{\Theta})$					1.53	117.86

Notes: Targeted SMM moments ($k = 1, \ldots, 30$), symbols from (S.D.3)–(S.D.9), empirical estimates $\widehat{m}_k$, standard errors $\widehat{\mathrm{SE}}_k$, inverse-variance weights $W_{kk} = 1/\widehat{\mathrm{SE}}_k^2$, and squared studentized residuals $q_k(\widehat{\Theta}) = \left[(\overline{m}_k(\widehat{\Theta}) - \widehat{m}_k)/\widehat{\mathrm{SE}}_k\right]^2$ under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$).

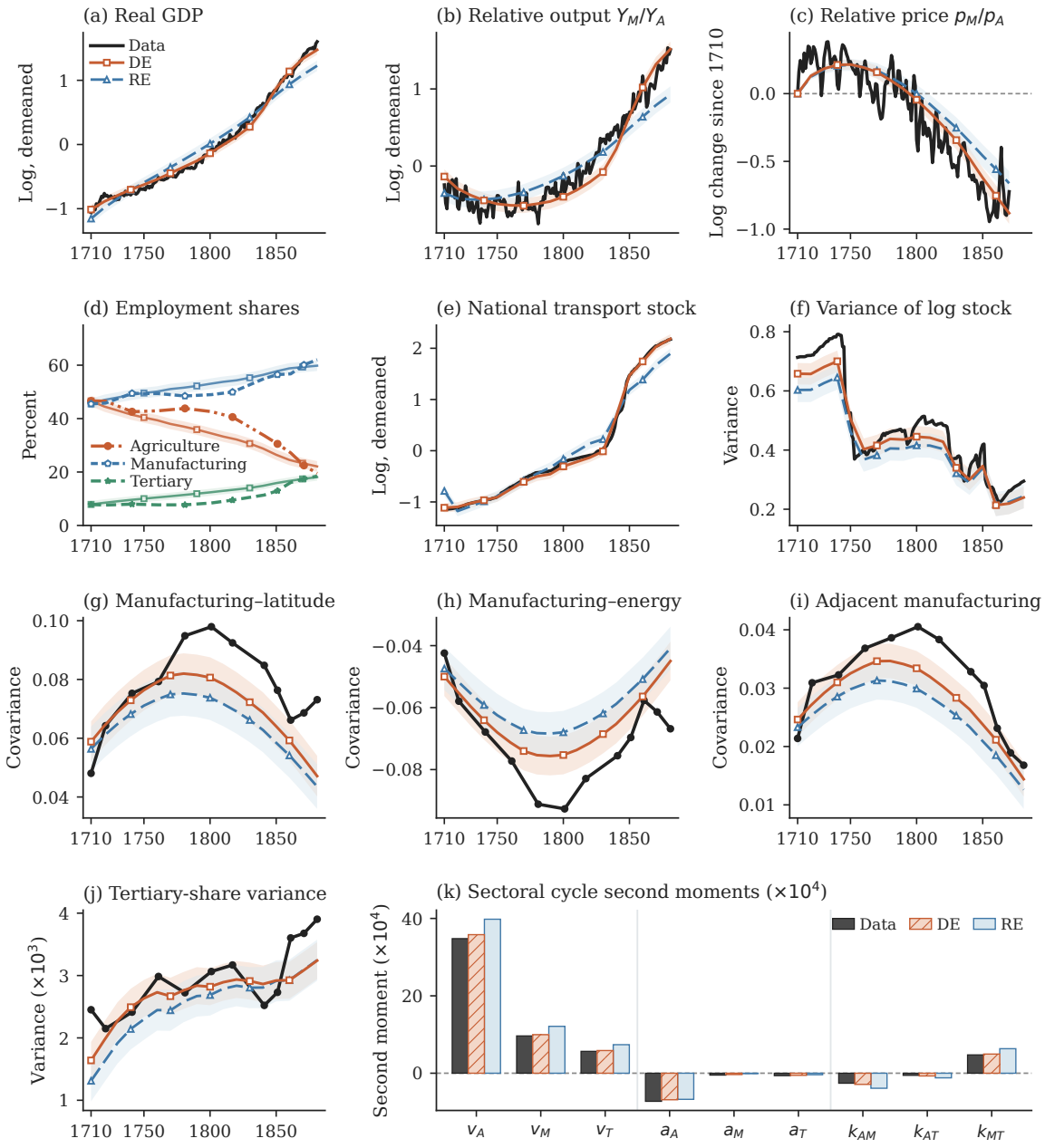


Figure S.D.1: Complete-Series Empirical Moment Profiles, 1710–1881

Notes: Historical profiles and simulated trajectories (means and 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and rational (RE, $\theta = 0$) expectations underlying Table 6. Panels (a)–(e) plot demeaned log real GDP and Y_M/Y_A, log change in p_M/p_A since 1710, occupational shares at the twelve reconstruction dates, and demeaned log transport stock (75); Panels (f)–(k) plot cross-band log-stock variance $V_{T,y}$ (S.D.5), manufacturing spatial covariances $(C_{l,y}, C_{e,y}, C_{N,y})$ (S.D.4), tertiary spatial variance $V_{T,y}^{\text{occ}}$ $(\times 10^3)$, and HP-filtered sectoral cycle variances, autocovariances, and cross-covariances $(v_s, a_s, k_{ss'})$ $(\times 10^4)$ (S.D.9).

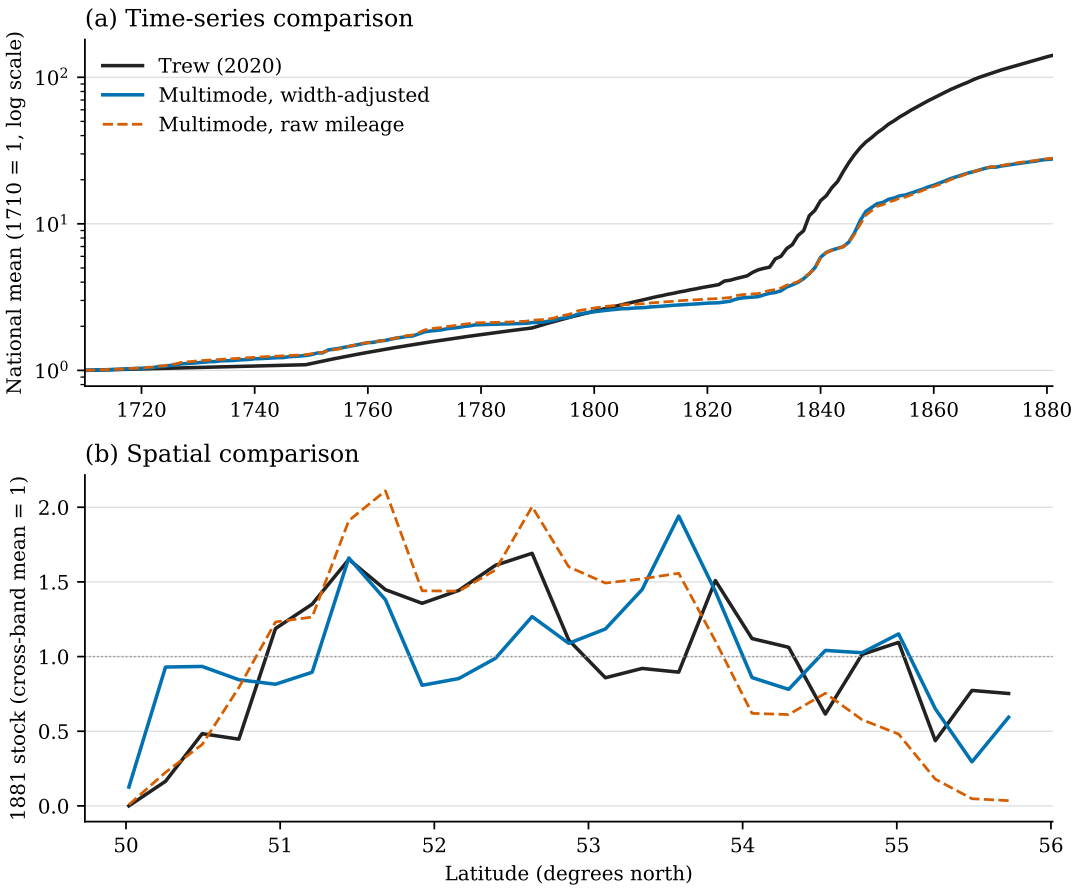


Figure S.D.2: Alternative Constructions of the Multimode Transport Stock

Notes: Panel (a) reports national mean transport stocks normalized to one in 1710. Panel (b) reports 1881 band-level stocks normalized by each series' cross-band mean, with and without the east–west land-width adjustment.

S.E Identification of the Structural Moment Map

Identification proceeds in four steps: (i) single-valuedness of the population moment map from Propositions 6 and 7, (ii) the factor-payment restriction linking fixed production elasticities to payroll shares and the manufacturing wedge, (iii) local point identification of $\boldsymbol{\psi}$ from the Jacobian rank and Hessian conditions of Rothenberg (1971), Iskrev (2010), Komunjer and Ng (2011), and Andrews et al. (2017), and (iv) analytical separation of θ from (m_I, χ_I) and $\boldsymbol{\rho}$.

S.E.1 Normalizations and observational equivalence

Fix the unit normalizations $p_A(1,t) = 1$, $\bar{L}_{1710} = 100$, and $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$, and let $\mathcal{T} \subset \mathbb{R}^p$ denote the open parameter domain containing the full parameter vector $\boldsymbol{\psi}$ (either unrestricted with $\theta \in (0, \bar{\theta})$ or restricted to $\theta = 0$). For a complete primitive shock history $u \sim \mathsf{Q}$, let $\mathscr{Y}(\boldsymbol{\psi}, u)$ denote the unique equilibrium trajectory of observables and $\Psi(\mathscr{Y}(\boldsymbol{\psi}, u)) \in \mathbb{R}^{d_m}$ the retained moment

vector. The population moment map and identified set at $m_0 \in \mathbb{R}^{d_m}$ are

$$m(\boldsymbol{\psi}) = \int \Psi(\mathscr{Y}(\boldsymbol{\psi}, u))\, d\mathsf{Q}(u), \tag{S.E.1}$$

$$\mathcal{I}(m_0) = \{\boldsymbol{\psi} \in \mathcal{T} : m(\boldsymbol{\psi}) = m_0\}. \tag{S.E.2}$$

A parameter vector $\boldsymbol{\psi}_0 \in \mathcal{T}$ is locally point identified if there exists an open neighborhood $\mathcal{N} \ni \boldsymbol{\psi}_0$ such that $\mathcal{I}(m(\boldsymbol{\psi}_0)) \cap \mathcal{N} = \{\boldsymbol{\psi}_0\}$.

PROPOSITION S.E.1: *Suppose the conditions of Propositions 6 and 7 (and Corollary A.1) hold at every state reached from the initial condition for each $\boldsymbol{\psi} \in \mathcal{T}$. If $\Psi(\mathscr{Y}(\boldsymbol{\psi}, u))$ is Q-integrable, then the population moment map (S.E.1) is single valued on $\mathcal{T}$.*

Proof. By Propositions 6 and 7 and Corollary A.1, each primitive shock history u generates a unique measurable outcome path $\mathscr{Y}(\boldsymbol{\psi}, u)$, whose Q-integral defines a unique moment vector $m(\boldsymbol{\psi})$. □

S.E.2 Direct restrictions and local rank

PROPOSITION S.E.2: *For every $\boldsymbol{\psi} \in \mathcal{T}$, the conditional hiring rule (29) implies that expected operating payroll relative to expected net producer revenue at active sites satisfies*

$$\frac{w(\ell,t)L_A(\ell,t)}{\mathbb{E}_t[P_A(\ell,t)_+ Y_A(\ell,t)]} = \alpha_A, \qquad \frac{w(\ell,t)L_M(\ell,t)}{\mathbb{E}_t[P_M(\ell,t)_+ Y_M(\ell,t)]} = \frac{\alpha_M}{1+\bar{\psi}_M \omega_M(\ell)}, \tag{S.E.3}$$

for all $\phi_{it}^*(\ell) \in [0,1)$ (and holds statewise whenever $P_i(\ell,t,\omega) > 0$, $\phi_{it}^*(\ell) = 0$, and spot prices and tolls are deterministic).

Proof. Since $Y_i(\ell,\omega) = Z_i(\ell)G_i(\ell,\omega)L_i^*(\ell)^{\alpha_i}$ whenever $P_i(\ell,\omega) > 0$, (29) gives

$$\mathbb{E}_t\big[P_i(\ell,\omega)_+ Y_i(\ell,\omega)\big] = Z_i(\ell) m_i(\phi_i^*(\ell);\ell) L_i^*(\ell)^{\alpha_i} = \frac{1+\psi_i(\ell)}{\alpha_i} w(\ell) L_i^*(\ell),$$

yielding (S.E.3). □

THEOREM S.E.1: *Fix an interior $\boldsymbol{\psi}_0 \in \mathcal{T}$. Suppose Proposition S.E.1 holds and m is C^1 on a neighborhood of $\boldsymbol{\psi}_0$. If the population Jacobian satisfies*

$$D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0) \in \mathbb{R}^{d_m \times p} \quad \textit{with} \quad \operatorname{rank} D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0) = p, \tag{S.E.4}$$

then $\boldsymbol{\psi}_0$ is locally point identified by the retained moments, and the rank is invariant to nonsingular whitening and smooth interior reparameterizations.

Proof. By (S.E.4), $D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0)$ contains a nonsingular $p \times p$ minor, and applying the inverse function theorem to the corresponding p-moment subvector yields local injectivity around $\boldsymbol{\psi}_0$. □

S.E.3 Pseudo-true values and branch regularity

Let $g(\boldsymbol{\psi}) = m^{\text{data}} - m(\boldsymbol{\psi})$ and $Q(\boldsymbol{\psi}) = g(\boldsymbol{\psi})' W g(\boldsymbol{\psi})$ for positive-definite W.

THEOREM S.E.2: *Fix an interior point $\boldsymbol{\psi}^*$ where g is C^2 and satisfies first-order stationarity*

$$2Dg(\boldsymbol{\psi}^*)' W g(\boldsymbol{\psi}^*) = 0. \tag{S.E.5}$$

If the exact criterion Hessian

$$H(\boldsymbol{\psi}^*) = 2Dg(\boldsymbol{\psi}^*)'WDg(\boldsymbol{\psi}^*) + 2\sum_{k=1}^{d_m}\left[Wg(\boldsymbol{\psi}^*)\right]_k \nabla^2 g_k(\boldsymbol{\psi}^*) \tag{S.E.6}$$

is positive definite, then $\boldsymbol{\psi}^*$ is a strict local minimizer of Q in the global pseudo-true set

$$\mathcal{P} = \arg\min_{\boldsymbol{\psi}\in\mathcal{T}} g(\boldsymbol{\psi})'Wg(\boldsymbol{\psi}), \tag{S.E.7}$$

and at exact fit ($g(\boldsymbol{\psi}^*) = 0$) the rank condition (S.E.4) suffices for $H(\boldsymbol{\psi}^*) \succ 0$.

Proof. Equations (S.E.5)–(S.E.6) are the gradient and Hessian of $Q(\boldsymbol{\psi})$, so $H(\boldsymbol{\psi}^*) \succ 0$ yields strict local convexity around $\boldsymbol{\psi}^*$; at $g(\boldsymbol{\psi}^*) = 0$, $H(\boldsymbol{\psi}^*) = 2Dg(\boldsymbol{\psi}^*)'WDg(\boldsymbol{\psi}^*) \succ 0$ if and only if $\operatorname{rank} Dg(\boldsymbol{\psi}^*) = p$. □

S.E.4 Joint identification of diagnosticity θ

Separate identification of θ from the construction cost parameters (m_I, χ_I) and the aggregate shock persistence vector $\boldsymbol{\rho}$ follows from the block-triangular structure of the moment map across Panels B, D, and E of Table 6 and the contrasting signs of their partial derivatives on investment–output comovement b_0 and first-order investment autocorrelation ρ_1. Linearizing the interior branch of the clipped investment rule (61), $x^{\theta}_{jt} = \chi_I^{-1}[\tilde{q}^{\theta}_{jt}/(m_I D^I_{jt}) - 1]$, around the deterministic transition path under the one-period diagnostic reference lag ($J_{\mathrm{DE}} = 1$, $\bar{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t]$) shows that a date-$t$ aggregate innovation $\boldsymbol{\varepsilon}_t$ shifts perceived date-$(t+1)$ productivity by $(1+\theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, whereas at date $t+1$ (under $\mathbb{E}_t[\boldsymbol{\varepsilon}_{t+1}] = \mathbf{0}$) the reference forecast updates to $\boldsymbol{\rho}\mathbf{z}_t$ and the date-t diagnostic belief wedge $+\theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$ vanishes while the elevated stock $\overline{T}_{t+1}$ lowers marginal continuation value. By contrast, under $\theta = 0$, lowering adjustment convexity χ_I (raising $-\chi_I$) or raising shock persistence ρ_M increases both b_0 and ρ_1:

$$\begin{gathered}
\frac{\partial b_0}{\partial \theta} > 0, \qquad \frac{\partial \rho_1}{\partial \theta} < 0, \qquad \frac{\partial b_{\mathrm{lead}}}{\partial \theta} < 0, \\
\frac{\partial b_0}{\partial(-\chi_I)}\bigg|_{\theta=0} > 0, \qquad \frac{\partial \rho_1}{\partial(-\chi_I)}\bigg|_{\theta=0} > 0, \qquad \frac{\partial b_0}{\partial \rho_M}\bigg|_{\theta=0} > 0, \qquad \frac{\partial \rho_1}{\partial \rho_M}\bigg|_{\theta=0} > 0.
\end{gathered} \tag{S.E.8}$$

Because the sectoral output variances and first-order autocovariances (v_s, a_s) in Panel D pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$ and the long-run linear trend of log transport stock in Panel B pins down the construction cost scale m_I, no movement in $-\chi_I$ can simultaneously raise b_0 and lower ρ_1, separating θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Full column rank of $D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0)$ in (S.E.4) is verified numerically at the estimates in Table 5. Table S.E.1 verifies that the fit advantage of diagnostic expectations is not driven by an extra free parameter, moment-block weighting, or literal-specification assumptions: Panel A re-estimates all $p = 25$ remaining structural parameters at fixed values $\theta \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ and compares each restricted specification against $\theta = 0$ via the misspecification-robust Rivers and Vuong (2002) equal-parameter z-statistic; Panel B reports the Andrews et al. (2017) local identifying information shares across moment blocks—showing that the investment-dynamics and transport-network blocks (Panels E and B) supply 85.7 percent (49.8 and 35.9 percent) of the identifying information for $\widehat{\theta}$—and confirms that $\widehat{\theta}$ remains stable across leave-one-block-out estimations ($\widehat{\theta} \in [0.666, 0.747]$, holding the nuisance block identified primarily by the omitted moment panel at its baseline estimate while θ, m_I, χ_I, and the remaining parameters are re-estimated so that $d_m > p_{\mathrm{sub}}$ in every sub-sample) as well as under equal-block studentized weighting ($\widehat{\theta}_{\mathrm{eq}} = 0.748$, SE $= 0.104$).

Table S.E.1: Restricted SMM Profile Across Fixed θ and Leave-One-Block-Out Estimates

Specification / statistic	$\theta = 0.0$ (RE)	$\theta = 0.3$	$\theta = 0.6$	$\theta = 0.9$	$\theta = 1.2$
Panel A. Restricted SMM estimation across fixed θ ($p = 25$ free parameters)					
Estimated free parameters p	25	25	25	25	25
Sub-criterion: Panels A–C ($Q_{\mathrm{A–C}}$, 14 moments)	41.74	15.37	2.28	2.16	14.70
Sub-criterion: Panel D (Q_{D}, 9 moments)	14.16	4.88	0.48	0.61	4.91
Sub-criterion: Panel E (Q_{E}, 7 moments)	61.96	20.81	2.43	2.77	18.60
SMM criterion, $Q(\widehat{\Theta}(\theta))$ (30 moments)	117.86	41.06	5.19	5.54	38.21
Rivers–Vuong (2002) z-statistic vs. $\theta = 0$	—	8.56	6.42	4.35	2.36
Rivers–Vuong (2002) p-value vs. $\theta = 0$	—	[< 0.0001]	[< 0.0001]	[< 0.0001]	[0.0184]
Adjustment cost convexity $\widehat{\chi}_I(\theta)$	1.589	1.728	1.867	2.006	2.145
Construction cost scale $\widehat{m}_I(\theta)$	0.020	0.021	0.022	0.023	0.024
Manufacturing AR(1) persistence $\widehat{\rho}_M(\theta)$	0.561	0.550	0.538	0.526	0.515
Manufacturing innovation s.d. $\widehat{\sigma}_M(\theta)$	0.042	0.041	0.039	0.038	0.037
Technological catch-up rate $\widehat{\lambda}_0(\theta)$	0.175	0.169	0.163	0.157	0.150
Iceberg transport cost scale $\widehat{\kappa}_0(\theta)$	0.073	0.076	0.079	0.082	0.085
Panel B. Moment-block information shares and leave-one-block-out estimates of $\widehat{\theta}$					
Moment block	Panel A	Panel B	Panel C	Panel D	Panel E lags
Andrews–Gentzkow–Shapiro (2017) info. share (%)	1.3%	35.9%	0.3%	12.7%	49.8%
Leave-one-block-out $\widehat{\theta}$ (SE)	0.739 (0.090)	0.723 (0.095)	0.739 (0.090)	0.747 (0.091)	0.666 (0.166)
Sub-sample criterion $Q_{\mathrm{sub}}(\widehat{\Theta})$	1.22	1.44	1.27	1.05	1.06

Notes: Panel A reports restricted SMM estimates fixing $\theta \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ ($p = 25$ free parameters) with misspecification-robust Rivers and Vuong (2002) equal-fit tests against $\theta = 0$ (p-values in brackets). Panel B reports Andrews et al. (2017) local identifying information shares for $\widehat{\theta}$ alongside leave-one-block-out estimates (standard errors in parentheses; equal-block studentized weighting yields $\widehat{\theta}_{\mathrm{eq}} = 0.748$, SE $= 0.104$).